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Artificial Intelligence: Arguments for Catastrophic Risk

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Abstract

Recent progress in artificial intelligence (AI) has drawn attention to the technology's transformative potential, including what some see as its prospects for causing large-scale harm. We review two influential arguments purporting to show how AI could pose catastrophic risks. The first argument — the *Problem of Power-Seeking* — claims that, under certain assumptions, advanced AI systems are likely to engage in dangerous power-seeking behavior in pursuit of their goals. We review reasons for thinking that AI systems might seek power, that they might obtain it, that this could lead to catastrophe, and that we might build and deploy such systems anyway. The second argument claims that the development of human-level AI will unlock rapid further progress, culminating in AI systems far more capable than any human — this is the *Singularity Hypothesis*. Power-seeking behavior on the part of such systems might be particularly dangerous. We discuss a variety of objections to both arguments and conclude by assessing the state of the debate.

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All authors contributed equally to this study.

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1 | INTRODUCTION

Artificial intelligence (AI) has arrived. Today's AI systems display various impressive capabilities: they handily defeat the best humans at games of strategy (Schrittwieser et al., 2019), score highly on standardized tests (OpenAI, 2023b), help design novel drugs (Bran et al., 2023; Jumper et al., 2021), create dazzling images via models like Dall-E 3, write and debug code via models like Github Codex, and converse compellingly with humans via models like GPT-4.

These capabilities are likely to progress rapidly in coming years as competition intensifies and hardware and algorithms improve. Some leading AI companies explicitly aim at creating general-purpose human-level AI.[1] Assuming this goal is achieved, developers will face strategic and economic incentives not to stop at human-level capabilities. We may soon[2] find ourselves sharing the world with a new kind of highly capable thinking thing, trained but not necessarily understood or fully controlled by us.

A growing number of philosophers and others fear that future AI technology might pose a catastrophic risk to humanity (Bostrom, 2014; Ord, 2020; Vold & Harris, 2021). In this article, we explore some of the arguments behind such worries.[3] In contrast with discussion of pandemics, nuclear weapons, and other global threats, the AI risk debate turns on distinctive issues in decision theory, ethics, and philosophy of mind. For this reason, the debate has been shaped by philosophers from its beginnings.

Our coverage here is limited in several ways. First, we primarily discuss arguments for *catastrophic* risk, involving globally devastating harm or threats to human existence, autonomy or potential. (This isn't to deny that major non-catastrophic harms from AI are also worth taking seriously.) Second, we focus on risk claimed to be *intrinsic* to certain kinds of AI systems, rather than risk associated with human error or malicious misuse.[4] (This isn't to say that the latter kinds of risks are smaller, but they won't be our focus here.) Finally, lacking space to cover every form of risk in detail, we concentrate on one widely discussed idea: the notion that some advanced AI systems are likely to function as agents pursuing goals, and as a result, are likely to engage in dangerous resource-acquiring, shutdown-avoiding, and correction-resisting behavior.

Section 2 outlines this *Problem of Power-Seeking*. Section 3 then discusses two claims about AI goals (the *Instrumental Convergence Thesis* and the *Orthogonality Thesis*) that are relied upon in making the case for the Problem of Power-Seeking. Assessing these claims is crucial for assessing the broader argument. In section 4, we turn to the *Singularity Hypothesis*, according to which rapid improvements, culminating in capabilities far beyond our species' level, will follow on the heels of human-level AI. If this hypothesis is true, it provides one reason to expect the sort of supercapable systems that can lead to the Problem of Power-Seeking. So assessing this hypothesis helps us to determine how concerned we should be by power-seeking AI systems.

It's worth noting that there's little consensus about the likelihood of catastrophe from the sources we discuss. A small but vocal contingent of pessimists takes human extinction to be the default outcome of advanced AI (cf. Yudkowsky, 2023). Other theorists take catastrophe to be unlikely but worth taking seriously, on the grounds that even small chances of disaster should be understood in detail and reduced if possible. (For instance, Carlsmith (2022) estimates that an AI catastrophe is >10% likely by 2070.) Still others question whether the risks rise even to this level of significance (Thorstad, 2022; §6.2, Thorstad, 2023). In general, the reader would do well to view the arguments below as aiming at plausibility rather than very high probability or proof.

2 | THE PROBLEM OF POWER-SEEKING

Existing AI systems have already caused significant harms. Pedestrians have been killed by self-driving cars, for instance, and AI-generated images have aided incendiary misinformation campaigns (Klee & Ramirez, 2023; Sanger & Myers, 2023). However, while such outcomes are tragic, they don't constitute catastrophes on the scale

discussed above. To pose such a risk, AI systems would likely need much greater power to affect the world. One prominent worry suggests that this is just what might happen: under plausible assumptions, some AI systems will seek power, successfully obtain it and go awry, potentially bringing about catastrophe (Yudkowsky, 2008; Omohundro, 2008; Bostrom, 2012, 2014, Carlsmith, 2022, forthcoming). This is the *Problem of Power-Seeking*.

We can spell out the case for this concern by considering four questions.

First, **why might AI systems seek power?** The standard argument appeals to the *Instrumental Convergence Thesis*, which in rough terms holds that certain subgoals are useful for achieving a wide variety of final goals (Bostrom, 2012; Omohundro, 2008). Such subgoals might include self-preservation, self-improvement, and resource acquisition, among others. The Instrumental Convergence Thesis seems to suggest that many AI systems will develop these subgoals, which we can describe as power-seeking subgoals, and so suggests that AI systems will seek power. We discuss this thesis further in §3.

Second, **why might power-seeking AI successfully acquire power?** One reason is that some expect us to develop *superintelligent AIs* with radically greater cognitive powers than the smartest humans. Such systems might be able to engage in sophisticated misinformation campaigns, develop new military technologies, profit from the stock market, and so on. Given such advantages, if superintelligent AI systems seek power, some will plausibly acquire it (Bostrom, 2014).

Despite the terminology, this argument doesn't rely on a unified notion of general intelligence. Nor does it assume that AIs will become conscious, acquire personhood, or possess humanlike beliefs and desires. Instead, what matters is that the systems are highly capable at various tasks that are useful for gaining power (Carlsmith, forthcoming). A more apt label for this notion might be *supercapability*.

Supercapability provides one reason AI systems might acquire power; there are at least two others. One is *supernumerosity*: since copying software is cheap and easy, the population of capable AIs could quickly grow large. If these systems cooperated, sheer numbers might allow them to seize substantial collective power (Karnofsky, 2022). Further, humans might voluntarily relinquish significant power to increasingly capable systems, because these systems can carry out useful tasks. For example, militaries might hand control of drones to AI.

Third, **why might AI systems acquiring power lead to catastrophe?** If AI systems seek power on large scales, this might lead to conflict with humans as competition arises for resources and influence. For example, AIs might seek to disempower humans to limit our interference with their goals and might seize control of resources we rely on in order to promote their own ends (Bostrom, 2014, p. 141). If these systems have acquired sufficient power, the outcomes of such behaviors could be catastrophic for humans.

Fourth, **why would we develop and deploy AI systems that posed catastrophic risks?** For a start, even if these risks were real and there was mounting evidence for this, some people might not take the danger seriously, just as some are unconcerned by climate change. Further, if AI proves useful and economically valuable, competition between AI companies and between governments will likely ensue (Carlsmith, 2022, §5). Competitive environments encourage rapid progress that might not involve adequate attention to safety. Finally, there's the possibility of *deceptive alignment*, which involves systems appearing safe during development but becoming dangerous when they're deployed in the world (Ngo et al., 2023, §4.2; Bostrom, 2014, pp. 142–145). In such cases, we might not realize the risk until it's too late.

This concludes our overview of the Problem of Power-Seeking. We now consider two objections.

First, the above arguments envisage AI systems with at least human-level cognitive capabilities that behave in ways that strike us as capricious and morally alien, pursuing unlimited power in the service of arbitrary goals. This is surprising. Perhaps it's also implausible. As AI systems develop cognitive capabilities at a similar level to our own, it's possible their motivations will also converge on something like ours (see Müller & Cannon, 2021). Catastrophic power-seeking might then be rare. Against this line of thought, *the Orthogonality Thesis* holds that arbitrarily high levels of intelligence can be combined with more or less any final goals (Bostrom, 2012). If this were true, a system could be superintelligent without being driven by concerns that are conducive to human flourishing. We discuss these issues further in §3.

According to a second objection, power-seeking worries ignore the fact that humans create AI systems and so control their form. We can design systems which won't cause large-scale harm, and we can implement regulations to ensure dangerous systems aren't deployed (cf. Pinker, 2015). Indeed, safety concerns and economic incentives point in the same direction here: unleashing catastrophe on the world would be bad for business, so even profit-driven companies will be motivated to invest in safety.

These suggestions are reasonable in principle, but it's unclear that they'll be easy to implement in practice. Ensuring that AI systems do what we want requires ruling out misaligned behavior contrary to designers' intentions.[5] If misaligned behavior proves difficult to foresee prior to deployment, as some suspect it will, then well-meaning developers and regulators may lack the tools to avert catastrophe.[6]

This *alignment problem* is at the core of many worries about catastrophic risk from AI. To get a sense of why it might prove difficult for designers to foresee and rule out misaligned behavior, consider the story of King Midas—or stories of golems and genies—where a simple wish is fulfilled in unexpected and undesirable ways. Part of the point of these stories is that wishes, when interpreted literally and without reference to our broader assumptions and desires, can easily go awry in ways we didn't anticipate. A supercapable AI system whose motives or understanding of the world differ even slightly from ours might resemble such dangerous wish-fulfillers (Muehlhauser & Helm, 2012).

More concretely, we can see how misalignment might arise by considering AI systems developed via *reinforcement learning*, a central technique underpinning contemporary AI. In reinforcement learning, a designer doesn't specify how a system is to carry out a given task; instead, the system learns how to do so via a training process. This involves the system being run on a task and then receiving feedback on how it performed, in the form of a reward signal. The system is then modified, based on this reward, so that it will better carry out the task. With enough iterations, this process can lead to highly competent systems.

Given this approach, there are at least two ways that misaligned behavior can arise: reward misspecification and goal misgeneralization (Ngo et al., 2023). *Reward misspecification* involves a system accidentally being rewarded for undesirable behavior (Krakovna et al., 2020).[7] This happens because it can be hard to specify a reward that captures exactly the desired behavior. For example, one system designed to win a boat racing video game was rewarded for achieving in-game points, a seemingly reasonable choice (Clark & Amodei, 2016). However, rather than winning, the resulting system steered the boat in circles in a way that allowed it to continually acquire points. Here, the reward was misspecified.

Even in the absence of misspecification, *goal misgeneralization* can arise (Langosco et al., 2022; Shah et al., 2022). This phenomenon arises because, in complex environments, reinforcement learning systems cannot be trained on every situation they might face. This raises the possibility that any goals developed might lead to desirable behavior in the situations encountered in training but lead to undesirable behavior in novel situations. This is goal misgeneralization. For example, Shah et al. (2022) trained an AI system to collect virtual apples and avoid being attacked by monsters. The system learned to collect shields, which helped with the latter task. Yet it continued to collect shields even when placed in environments without monsters, where acquiring shields was a waste of effort. This system's behavior generalized poorly to novel situations.

Reward misspecification and goal misgeneralization represent two ways that AI systems could develop unwanted and potentially dangerous behavioral tendencies. These might be difficult to anticipate prior to deployment, so safety-minded intentions might not suffice to prevent harm.

To summarize: AI risk theorists maintain that we have grounds to think AI systems might seek and acquire power in a way that leads to catastrophe, and grounds to think we might deploy such systems anyway. This is the Problem of Power-Seeking.

3 | AI GOALS

The above argument appealed to two claims relevant to the nature of AI goals: the Instrumental Convergence Thesis and the Orthogonality Thesis. We'll now examine these.

We start with the Instrumental Convergence Thesis, the claim that certain resource-acquiring, self-improving and shutdown-resisting subgoals are useful for achieving a wide variety of final goals. Given this thesis, we might expect sufficiently advanced goal-directed AI systems to pursue such subgoals.

Instrumental Convergence is motivated by the observation that agents possessing more material resources, foresight, factual knowledge, and persistent influence on the world are typically better positioned to get what they want in many domains (Bostrom, 2012; Omohundro, 2008).[8] For example, acquisition of resources like money is useful for achieving many goals. Consider an AI system with the goal of proving mathematical theorems. This system could use money to hire research assistants or to pay for computing infrastructure with which to search the space of possible proofs. Similar motivations can be given for the remaining subgoals.[9]

The subgoals comprising the Instrumental Convergence Thesis form a heterogeneous group; they need not stand or fall as genuinely universal together. Accordingly, some skeptics have focused their criticisms on particular members of the set. Salib (forthcoming) argues against self-improvement as a convergent subgoal, on the grounds that unimproved AIs may be unable to control the goals of their improved successors, and may therefore view self-improvement as a risk to be avoided. Meanwhile Gallow (2023) evaluates the case for various potential subgoals, reaching the mixed conclusion that some are plausibly instrumentally convergent while others are not (see also Goertzel, 2015).

Alternatively, one might raise a wholesale objection to the Instrumental Convergence Thesis, as Goertzel does when he suggests that the thesis's intuitive plausibility relies on a tendency to anthropomorphize (Goertzel, 2015).[10]

We turn now to the Orthogonality Thesis: the claim that arbitrarily high levels of intelligence can be combined with almost any final goal. Bostrom's case for this thesis rests on a conception of intelligence as skill at instrumental reasoning, or "search[ing] for instrumentally optimal plans and policies" (2012, 73).

Müller and Cannon (2021) accept that Orthogonality might hold for intelligence in this instrumental sense. They argue, however, that AI systems would pose a threat to humanity only if they possessed a richer sort of general intelligence, consisting partly in the ability to reflect on and modify one's goals. According to Müller and Cannon, the Orthogonality Thesis is false if interpreted in terms of this latter notion of intelligence, because a highly capable generally intelligent system would seek rational and ethical coherence (and so might not retain any arbitrary initial goal). Of course, both of these claims rest on controversial assumptions. As an empirical matter, history seems to offer vivid illustrations of actors threatening large subsets of humanity without undergoing the kind of transformative self-examination Müller and Cannon describe; if this combination of general intelligence and dangerous goals is common for us, why think it's unlikely for AI systems?

In addition to arguments targeting each thesis in isolation, there are also holistic considerations relevant to both. We'll discuss two of these.

First, Goertzel (2015) notes that questions of safety will be decided by the specific goals that future AI systems actually develop. These will likely be constrained—by human intentions, technological limitations, training materials and procedures, and so on—rather than entirely arbitrary.

This observation might seem to challenge the relevance of both theses. Ultimately, what matters is not whether subgoals are useful for achieving *a wide variety of* final goals, but whether they're useful for achieving future systems' actual goals, which may or may not be typical of goals in general.[11] So Instrumental Convergence might be true but irrelevant for the real cases of interest in our world. Likewise, what matters is not whether high levels of intelligence can be combined with *most* final goals, but whether intelligence is compatible with future systems' actual goals. If not, Orthogonality will be similarly irrelevant.

These observations suggest that power-seeking arguments would be improved by paying greater attention to the particular types of goals that future AI systems are most likely to pursue. (The same advice also applies to some of these arguments' critics: for instance, Gallow's negative results on Instrumental Convergence assume that an AI system's desires are sampled randomly from the space of all possible desires.) On the other hand, if there are strong reasons to believe that the appearance of constraint is misleading and future AI goals might well be essentially arbitrary, these reasons deserve to be stated more clearly than they have been.

A second holistic point is that Instrumental Convergence and Orthogonality both rely on the assumption that AI systems will have goals (and goal-pursuing abilities) of some kind. This assumption may deserve further scrutiny.[12]

In its favor, Carlsmith (2022, §3) argues that goal-based architectures are very useful: systems that have goals and competently aim to achieve them will be well placed to carry out complex tasks in the real world.[13] If this is right, then human designers might deliberately give AI systems the structure needed for goals (as was tried, for instance, with Auto-GPT, an AI assistant that augments ChatGPT with internet access and various interactive capabilities).[14] Alternatively, the usefulness of goals might mean that, if we train a machine learning system to competently carry out a complex task, goal pursuit might emerge during training, even without us intentionally creating this structure (Carlsmith, 2022, §3.3, Ngo et al., 2023, §3.2). However, Drexler (2019) pushes back against these views, arguing that it is both possible and desirable to create advanced AI systems that don't have goals in the relevant sense.

A second class of arguments for expecting goal-pursuing AI systems are so-called coherence arguments (Omohundro, 2007; §3, Omohundro, 2008; Yudkowsky, 2019). In simple terms, these arguments hold that AI systems will be open to undesirable forms of manipulation unless they behave as *expected utility maximizers*, which can be viewed as certain sorts of goal-driven agents. Given the further claim that sufficiently advanced AI systems won't be open to undesirable manipulation, we get the conclusion that these systems will be expected utility maximizers. Hence, they'll have goals in a relevant sense. However, Thornley (2023) and Bales (forthcoming) have pushed back against these arguments, in part by noting that the connection between avoiding manipulation and maximizing expected utility is less straightforward than simple versions of the coherence argument assume.[15]

All of these debates—about the Instrumental Convergence Thesis, the Orthogonality Thesis, and whether AI will be goal-driven—strike us as live and worth further philosophical attention.

4 | THE SINGULARITY HYPOTHESIS

The Singularity Hypothesis is the hypothesis that there will be a period of rapid recursive improvement in the capabilities of AI systems following the point at which AI systems become able to contribute to research on AI. It bears on questions of catastrophic risk in so far as the result of such an "intelligence explosion" could be super-capable artificial systems which, if not aligned with human interests, might pose a threat to humanity.

Dialectically, the Singularity Hypothesis makes contact with the Problem of Power-Seeking at two points: first, misaligned agential AI systems with instrumentally convergent reasons to engage in self-improvement might trigger a singularity-type recursive process; second, human actors seeking to improve the capabilities of AI systems could initiate a singularity-type recursive process which gives rise to misaligned supercapable AI systems. Thus it is conceivable that an "intelligence explosion" could be either the *result* or the *cause* of power-seeking behavior in AI systems.

The two most extended treatments of the Singularity Hypothesis from a philosophical perspective are due to Chalmers (2010) and Bostrom (2014), and these will be our focus in what follows.[16]

Chalmers (2010) summarizes the case for the Singularity Hypothesis as follows:

1. There will be AI (before long, absent defeaters).
2. If there is AI, there will be AI+ (soon after, absent defeaters).
3. If there is AI+, there will be AI++ (soon after, absent defeaters).
4. So: There will be AI++ (before too long, absent defeaters).

Here, AI is understood as human-level artificial intelligence, AI+ is understood as artificial intelligence that surpasses most humans, and AI++ is understood as intelligence that is "far greater than human level" (2010, p. 11).

Chalmers motivates premise 1 by suggesting that if evolution could produce human-level intelligence, there can be no in-principle barrier to humans themselves producing the same level of intelligence artificially. The rapid advancement of the capabilities of AI systems over the past decade's "deep learning revolution" further bolsters the plausibility of this first premise.

When it comes to premise 2, Chalmers suggests that if we create AI using an "extendible method" — a method that could, if further developed, yield more intelligent systems — we should expect to be able to produce AI+ soon after. Not all methods of creating AI are plausibly extendible. As Chalmers points out, however, even if AI is not initially achieved using an extendible method, it could subsequently be used to discover an extendible method. This would suffice for the truth of premise 2.

The idea behind premise 3 is that the recursive improvement that will follow from the rise of AI+ will soon lead to AI++. Chalmers motivates the idea of recursive improvement by appealing to what he calls the *proportionality thesis*: the thesis that "increases in intelligence (or increases of a certain sort) always lead to proportionate increases in the capacity to design intelligent systems" (2010, 21). Given the proportionality thesis, once we reach an AI+ with the ability to design a system more intelligent than itself, we know that the system it designs will be able to design a system still *more* intelligent, and so on until AI++ is achieved.

Bostrom's (2014) version of the argument appeals to interplay between two factors: *optimization power* — the amount of effort applied to improving AI systems, weighted by its quality — and the *recalcitrance* of the problem of improving them.[17] When optimization power is great and recalcitrance is low, Bostrom suggests, we should expect AI systems to improve rapidly. On this way of framing the issue, the key question in considering the Singularity Hypothesis is how we should expect optimization power and recalcitrance to change as we move from existing AI systems toward supercapable ones.

Bostrom offers a number of considerations supporting the idea that recalcitrance is likely to remain constant or decrease in the future, while optimization power is likely to increase. Even if it proves difficult to improve the algorithms implemented by artificial systems, for example, Bostrom argues that this recalcitrance will likely be offset by the comparative ease of improving such systems by investing in more hardware and giving them access to more content. At the same time, we should expect a rapid increase in optimization power due to growing investment of resources like capital and human researchers and the fact that, as AI systems become increasingly capable, they themselves will be able to contribute increasing amounts of optimization power. Indeed, Bostrom suggests that after the event he calls "crossover" — the moment when most of the optimization power applied to the problem of improving AI systems comes from AI systems themselves — we should expect the capabilities of AI systems to undergo rapid improvement because "any increase in the system's capability [will translate] into a proportional increase in the amount of optimization power being applied to its further improvement" (2014, p. 74). Thus Bostrom, like Chalmers, thinks about the singularity through the lens of a proportionality thesis.[18]

Both Chalmers and Bostrom are aware that the fact that a process of recursive improvement is possible does not entail that it will actually occur. Chalmers considers two broad classes of circumstances which might prevent a singularity-type recursive process from resulting in superintelligent AI systems. First, *situational defeaters* include natural or manmade disasters, resource limitations, and other conditions which might interrupt the process of recursive improvement. Second, *motivational defeaters* include disinclination on the part of either humans or AI systems to initiate or maintain the process of recursive improvement, as well as active attempts to prevent or interrupt that process. Bostrom discusses similar possibilities in the context of exploring how recalcitrance might evolve in the future.

From the perspective of concern about catastrophic risks to humanity, many of the situational defeaters Chalmers considers may strike us as cold comfort. For example, a natural or manmade disaster severe enough to permanently arrest technological progress toward superintelligent AI would itself likely be a global catastrophe.

At the same time, there are reasons to think that the kinds of motivational defeaters Chalmers considers are unlikely permanently to prevent a singularity-type outcome. There are strong military and business incentives for state and corporate actors to develop increasingly sophisticated AI systems, as well as competitive pressures for

them to do so before anyone else. In the absence of any global organization with the power to enforce a permanent moratorium on AI development, disinclination and efforts at active prevention are likely to be local and temporary.

It is worth emphasizing that, as with the arguments discussed in §2 above, the ubiquitous language of intelligence in discussions of the Singularity Hypothesis is eliminable. As Chalmers shows, singularity-type arguments can be constructed for any capability that is either (i) self-amplifying in the sense that it obeys a version of the proportionality thesis, or (ii) correlated with a self-amplifying capability. For example, if the capability to design and train deep-learning systems is self-amplifying and the capability to do scientific research is correlated with the capability to design and train deep-learning systems, there are singularity-style arguments that we should expect to see an "explosion" of both capabilities after AI reaches human-level competence at designing and training deep-learning systems.

Given their radical conclusions, arguments for the Singularity Hypothesis have always attracted controversy. For example, Modis (2012), Plebe and Perconti (2012), and Thorstad (2022) argue on empirical grounds that we should expect the rapid growth in the capabilities of artificial systems observed in the past several decades to level off in the near future rather than continue.[19] Thorstad also engages directly with the work of Chalmers and Bostrom, arguing that Chalmers does not sufficiently motivate his proportionality thesis and that the reasons Bostrom provides for thinking that recalcitrance will remain low in the future are either unconvincing or do not motivate the idea of an intelligence explosion.

In our view, neither the proponents of singularity-type arguments nor their critics have made a fully convincing case. One important and undertheorized question is whether levels of intelligence, or of any given putative self-amplifying capability, have the right mathematical structure for proportionality theses to make sense. To say that the value of a function $C(t)$ (say, our capacity to design intelligent systems at time t) increases in proportion to the value of another function $I(t)$ (say, the amount of intelligence at our disposal at time t) is to say that there is a positive constant a such that $C(t) = a * I(t)$. The proportionality thesis therefore implies that "the amount of intelligence at our disposal" has something like the structure of the real numbers, so that multiplication by a constant is possible and the quantity $a * I(t)$ makes sense. But if levels of intelligence have no more mathematical structure than that of a total order, for example, we can't speak of $a * I(t)$: we may be able to say of two amounts of intelligence that the first is larger than the second, but not that the first is a times larger. Under these circumstances, then, it is not possible to formulate a proportionality thesis.

A related question is whether proportionality theses, if they are coherent, are plausible. While it seems clear that increases in a system's intelligence will not be accompanied by *reductions* in its capacity to design intelligent systems or the optimization power it is able to contribute to various problems, the claim that every increase in intelligence is accompanied by a proportionate increase in design capacity or optimization power is much stronger. Chalmers himself considers the possibility of "diminishing returns" in design capability as intelligence increases, which challenges premise 3 of his argument. Why think that the proportionality thesis is more likely to be true than the diminishing returns thesis?

While we sympathize with Thorstad's call for more evidence in support of the proportionality thesis, we also think this thesis is less outlandish than it may at first seem. As Chalmers formulates it, the proportionality thesis is that increases in intelligence *lead to* proportionate increases in design capability. But the way in which this leading-to occurs might be quite indirect. For example, increasing the intelligence of a given system by 10% might lead to a proportionate increase in design capability by giving human engineers ideas about how to improve the intelligence of a distinct system — one better at designing AI systems — by 10%. Nor do the increases posited by the proportionality thesis need to happen immediately or be deterministic. It could take years of chancy trial and error for a 10% increase in the intelligence of AI systems to translate into a 10% improvement in their design capabilities. And the proportionality thesis does not need to hold in full generality to support Chalmers's argument: all that is required is that it hold for a number of capability improvements sufficient to achieve AI++ (Chalmers, 2010, p. 22).

5 | CONCLUSION

Are increasingly capable AI systems developed in the future likely to seek power? Will progress in AI lead to the kind of recursive improvement described by the Singularity Hypothesis? While some theorists have reached strong conclusions in this area, these questions strike us as unsettled. Continued philosophical research must be done to assess the plausibility of arguments for catastrophic risk from AI systems.

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CONFLICT OF INTEREST STATEMENT

No conflicts of interest to declare.

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ENDNOTES

[1] E.g., Google's DeepMind: "Our long term aim is to solve intelligence, developing more general and capable problem-solving systems, known as artificial general intelligence" (https://www.deepmind.com/about). And OpenAI: "We believe our research will eventually lead to artificial general intelligence, a system that can solve human-level problems" (https://openai.com/research/overview). If this optimism among technical experts surprises some readers familiar with the skeptical arguments of Hubert Dreyfus (1965, 1972), it is worth pointing out that contemporary machine learning has moved beyond many of the paradigms against which Dreyfus's criticisms were leveled.

[2] How soon? In a large 2023 survey of published AI researchers, superhuman general-purpose AI was expected by 2047 with 50% likelihood (Grace et al., 2024). Meanwhile, as of January 2024, the forecasting site Metaculus gives 2032 as the aggregated expected date for human-level AI, based on over 2500 predictions (https://www.metaculus.com/questions/5121/date-of-artificial-general-intelligence/). For a comparison and synthesis of a number of estimated timelines, see Wynroe et al. (2023). Of course, predicting long-term technological developments is no trivial task even for experts, and the historical record of AI timeline estimates suggests that developments in the field may be especially difficult to foresee.

[3] A future companion article for Philosophy Compass by two of the present authors (D'Alessandro and Kirk-Giannini) will examine proposals for developing AI safely.

[4] For discussion of biochemical misuse risks, see D'Alessandro et al. (2023). For general taxonomies of AI risk, see Yampolskiy (2015), Zwetsloot and Dafoe (2019), Hendrycks et al. (2023).

[5] For discussions of the concept of alignment, see Gabriel (2020), Carlsmith (2022, §4.1).

[6] One can also question whether AI companies might not willingly take a gamble on catastrophic outcomes if they believed the risk to be sufficiently low and the potential upside sufficiently high. See Carlsmith (forthcoming), §6 for further discussion. Thanks to David Krueger for raising this issue.

[7] Some terminology: *reward hacking* and *specification gaming* refer to behaviors that take advantage of reward misspecification to achieve high reward in an undesirable way.

[8] These assets are less helpful for some goals, such as the goal of doing nothing. This is why Instrumental Convergence is framed in terms of a "wide variety" of final goals.

[9] For attempts to formally prove that various behaviors are instrumentally convergent, see Turner et al. (2021), Turner and Tadepalli (2022).

[10] Another possibility: humans might deliberately give AI systems power-seeking goals. If so then the problem of power-seeking doesn't rely on Instrumental Convergence.

[11] For work on whether present-day systems display the behaviors predicted by Instrumental Convergence, see Baker et al. (2020), OpenAI (2023a, §2.9).

[12] The question isn't whether human developers of AI will have goals or whether a reward signal will be used in training but whether systems will themselves have goals (Ngo et al., 2023, §3).

[13] Carlsmith's discussion of whether AI systems will be agentic planners is roughly equivalent to our question of whether these systems will have goals.

[14] There's currently a great deal of interest in agentized language models, which might suggest that systems with goals are likely to become the norm in the future. See, for example, Wang et al. (2023), Goldstein and Kirk-Giannini (2023).

[15] See also Drexler (2019), §6.

[16] Other sources include Good (1951, 1959, 1962, 1965), Moravec (1988), Vinge (1993), Bostrom (1998, 2003), Kurzweil (2005), Hanson (2008), Muehlhauser and Salamon (2012), and Loosemore and Goertzel (2012).

[17] Bostrom writes about optimization power and recalcitrance as though they are continuous quantities representable by the real numbers but is explicit that this way of framing things is only a heuristic (2014, p. 65). We believe the charitable way to read Bostrom is as committed to the weakest structural assumptions about intelligence, optimization power, and recalcitrance that make modeling future changes in the intelligence of AI systems using his equations predictively useful. Identifying what these structural assumptions are is an interesting problem for future work.

[18] While the arguments of Bostrom and Chalmers remain the most influential among academic philosophers, readers may also be interested in two recent works which adopt a "compute-centric framework" for estimating how long it will take AI systems to reach human-level capabilities (Cotra, 2020; Davidson, 2023). This approach begins by estimating how much computing power would be required to train a system with a given level of capability, and then constructs a model of when we're likely to use this much computing power to train a system. One way of thinking about compute-centric models which we find helpful is as spelling out in more detail how the relationship between optimization power and recalcitrance is likely to evolve in the future.

[19] See also Vold and Harris (2021) for a brief overview.

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